APPENDIX II

SÃO TOMÉ AND PRÍNCIPE: DEBT SUSTAINABILITY ANALYSIS FOR LOW-INCOME COUNTRY FRAMEWORK [1]

1. **São Tomé and Príncipe has a moderate risk of debt distress after debt relief under the HIPC Initiative and the Multilateral Debt Relief Initiative (MDRI)**. The results suggest that the attainment of debt relief at the HIPC completion point and MDRI debt relief will bring about a substantial improvement in the country's debt outlook, though debt ratios will continue to be vulnerable to shocks until oil production starts in 2012.[2] The debt outlook remains vulnerable even under the baseline scenario, as indicated by a breach of debt thresholds prior to the production of oil. The risk of debt distress increases significantly under a scenario with no oil production and no fiscal adjustment. Reducing the risk of debt distress will therefore depend critically on maintaining sound fiscal policy consistent with a prudent borrowing strategy for the medium and long term as well as policies oriented to achieving broad-based long-run economic growth.

A. BACKGROUND

2. **São Tomé and Príncipe's medium- and long-term external debt at the end of 2005 is estimated at US$325 million in nominal terms, corresponding to a NPV of debt-to-export ratio of 299 percent**. The share of multilateral debt is 61 percent, of which about 88 percent is owed to the IDA, African Development Bank, and IMF. The share of official bilateral debt is about 37 percent, of which 38 percent is owed to Paris Club creditors. The share of commercial debt is only 2 percent. São Tomé and Príncipe is current with its obligations to multilateral creditors and has cleared most of its arrears with Paris Club creditors as part of a debt rescheduling agreement on Cologne terms (September 2005). The authorities are currently in the process of signing bilateral agreements with Paris Club creditors and reconciling debt records with other creditors.

3. **The HIPC completion point is expected to be reached in early 2007**. The debt service-to-GDP ratio was about 6.5 percent of GDP in 2006 and is expected to be 3.6 percent in 2007—a result of both enhanced HIPC and MDRI debt relief. MDRI will result in

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[1] This appendix updates the debt sustainability analysis (DSA) prepared in February 2006 (Country Report No. 06/349, October 2006). This LIC DSA makes use of the updated debt data base which was reconciled for the HIPC completion point and also incorporates revised macroeconomic projections.

[2] São Tomé and Príncipe is currently classified as a "Poor Performer" in IDA's Country Policy and Institutional Assessment (CPIA) index. Under the joint IDA/IMF debt sustainability framework the thresholds are: 30 percent for the NPV of debt-to-GDP ratio, 100 percent for NPV of debt-to-exports ratio, and 15 percent for the debt service-to-exports ratio.

substantial debt service savings, reflecting the large share of debt to the IDA, AfDF, and IMF in the country's total debt.

São Tomé and Príncipe: External Public Debt
(Millions of U.S. dollars)

	1999	2005
Total	**292.9**	**324.6**
Multilateral institutions	168.1	200.2
IDA	60.9	72.1
AfDF	88.8	99.8
IMF	0.0	3.2
Other	18.3	25.2
Bilateral and commercial	120.0	119.7
Paris Club	51.8	45.4
Non-Paris Club	68.2	74.3
Commercial creditors	4.8	4.8

Sources: São Tomé and Príncipe authorities, World Bank and IMF staff estimates.

B. Medium-Term Macroeconomic Framework

4. **The baseline macroeconomic assumptions of the HIPC DSA and the LIC DSA are the same**.[3] The baseline scenario assumes HIPC completion point in early 2007 and full MDRI debt relief. GDP growth is expected to accelerate over the medium term to 8 percent and then slow gradually to 5 percent in the long term.[4] Domestic investment would rise sharply from about 35 percent of GDP to an average of 60 percent in the medium term, largely on account of oil sector investment initiatives. The rise in the investment ratio will require continuing reliance on foreign savings, although national savings are forecast to rise on account of increased private savings once stability and economic growth become firmly established.

[3] See Box 4 in the HIPC Completion Point Document.

[4] The baseline scenario excludes the proposed topping-up of HIPC relief at completion point. Debt ratios after topping-up of HIPC relief and MDRI are presented in an alternative scenario.

5. **Risks to the macro framework arise from a possible weakening of the fiscal consolidation effort, potential shortfalls in foreign assistance triggered by the expected oil boom, and lower or no oil production**. All scenarios assume no domestic public debt (no significant development of a domestic-based market for government debt is foreseen) and most of the scenarios assume external borrowing on concessional terms. In particular, the scenario of no oil production assumes no fiscal adjustment, as in the HIPC debt sustainability analysis. This scenario also assumes less concessional terms for external borrowing, reflecting the worsening of the fiscal stance.

C. Fiscal Debt Sustainability Analysis

6. **In the baseline scenario, São Tomé and Príncipe's debt outlook shows a significant improvement, but remains vulnerable until oil production starts** (Table 2 and Figure 1). Both the NPV of debt-to-GDP and NPV of debt-to-revenue ratios remain low throughout the projection period. The debt service-to-revenue ratio is projected to remain below 5 percent after 2007, declining sharply once oil production starts. Topping-up of HIPC assistance improves all debt ratios, particularly the debt service-to-revenue ratio in the 2007–12 period (Table 2).

7. **The sensitivity analysis shows that the debt indicators are particularly vulnerable to a deceleration in GDP and deterioration of the primary balance before oil production starts**. Under the most extreme stress test, which assumes lower real GDP growth and primary deficit (Bound test B3 in Table 2), São Tomé and Príncipe's NPV of public debt-to-revenue ratio increases substantially above the baseline scenario from 2007 to 2010, then falls sharply after oil exports start in 2012 (Figure 1 and Table 2). However, the debt service-to-revenue ratio increases only gradually, as a result of assuming borrowing on highly concessional terms, remaining below 10 percent throughout the projection period.

8. **Under a scenario with no oil production and no fiscal adjustment, the public debt dynamics become explosive** (alternative scenario A1 in Table 2, and Figure 1). In this scenario, all debt indicators are projected to increase rapidly. Both the NPV of debt-to-GDP and the NPV of debt-to-revenue ratios are projected to increase substantially throughout the projection period, leading to an increasing debt service-to-revenue ratio.

D. External Debt Sustainability Analysis

9. **The baseline scenario shows favorable external debt dynamics. Debt relief under the HIPC Initiative and MDRI allows the debt ratios to improve substantially, though ratios remain close to the policy-dependent thresholds until oil production starts** (Table 4 and Figure 2). The NPV of debt-to-GDP ratio is projected to remain below the 30 percent policy-dependent debt-burden threshold throughout the projection period. The NPV of debt-to-exports ratio is projected to increase gradually from 65 percent in 2006 to 105 percent by 2011, when it breaches the policy-dependent threshold, but then to fall sharply after 2012, when oil production starts. The debt service-to-exports ratio is projected

to decline from 12 percent in 2006 to 7 percent on average during 2007–11, before declining sharply once oil production starts.[5] Topping-up HIPC assistance improves all debt ratios, particularly the debt service-to-exports ratio (Figure 3).

10. **The sensitivity analysis shows that in the medium term the debt indicators are particularly vulnerable to a deceleration in GDP and export growth.** In the most extreme stress test, which assumes a two-year shock to the average growth of real GDP growth and export, to the GDP deflator denominated in U.S. dollar, and to the net non-debt-creating flows (Bound test B5 in Table 4), São Tomé and Príncipe's external debt indicators are projected to increase substantially above the policy-dependent thresholds from 2007 to 2010, then fall sharply after oil exports start in 2012 (Figure 3 and Table 4). The debt service-to-exports ratio also increases significantly, but remains below the 15 percent threshold and falls after oil production starts.

11. **In a scenario with no oil production and no fiscal adjustment, external debt vulnerability becomes acute** (alternative scenario A1 in Table 4 and Figure 2). In this scenario, all debt indicators are projected to increase rapidly. Both the NPV of debt-to-GDP and the NPV of debt-to-exports ratios are projected to remain above the policy-dependent threshold, leading to an ever increasing debt service-to-exports ratio in the absence of fiscal adjustment.

E. Conclusion

12. **São Tomé and Príncipe's risk of debt distress after debt relief under the HIPC Initiative and MDRI will be moderate, but it will remain vulnerable to developments in the oil sector.** Despite a significant reduction in the NPV of debt at the HIPC completion point, São Tomé and Príncipe's public debt will also be vulnerable to shocks to the exchange rate, exports, and foreign grants, at least until oil production starts in 2012. External debt vulnerability would be further exacerbated in the absence of sound macroeconomic policies and undue delays in oil production. These risks underline the need for prudent fiscal and debt management policies, continued reliance on concessional financing, and implementation of structural reforms essential to support broad-based long term economic growth.

[5] The debt ratios of the LIC DSA may differ from the HIPC DSA as they use different methodologies. The LIC uses the same-year exports, while HIPC DSA uses a three-year backward average. The LIC DSA uses a fixed 5 percent discount rate, while the HIPC DSA uses currency-specific discount rates. Finally, the LIC DSA debt service projections use WEO exchange rates, while the HIPC DSA uses fixed exchange rates (end-2005).

Table 1.São Tomé and Príncipe: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005–26
(Percent of GDP, unless otherwise indicated)

	Actual			Estimate	Projections									
	2005	Historical Average 5/	Standard Deviation 5/	2006	2007	2008	2009	2010	2011	2006–11 Average	2012	2013	2016	2026	2012–26 Average
Public sector debt 1/	508.8			438.2	78.6	72.0	64.7	58.3	51.5		12.8	4.9	3.6	4.7	
Of which: foreign-currency denominated	508.8			438.2	78.6	72.0	64.7	58.3	51.5		12.8	4.9	3.6	4.7	
Change in public sector debt	77.0			-70.6	-359.6	-6.6	-7.2	-6.4	-6.9		-38.7	-7.9	0.2	0.0	
Identified debt-creating flows	-37.9			-75.2	-52.5	2.4	-3.6	-2.7	-4.4		-38.4	-7.8	0.3	-0.4	
Primary deficit	-28.1	9.2	15.0	-3.1	4.5	9.4	3.5	3.4	2.0	3.3	-0.2	-0.1	0.0	-0.1	-0.1
Revenue and grants	92.8			73.2	58.1	51.9	55.5	53.3	55.2		15.2	7.5	16.3	43.9	
Of which : grants	26.7			22.6	19.0	19.0	18.6	18.1	16.3		4.2	1.5	0.1	0.0	
Primary (noninterest) expenditure	64.7			70.1	62.6	61.3	59.0	56.6	57.1		15.0	7.4	16.3	43.8	
Automatic debt dynamics	-5.0			-67.8	-55.1	-5.4	-5.6	-4.7	-5.1		-37.9	-7.6	0.5	0.1	
Contribution from interest rate/growth differential	-31.5			-45.5	-33.6	-4.3	-4.3	-4.2	-3.8		-38.1	-7.7	0.4	0.0	
Of which : contribution from average real interest rate	-7.1			-7.8	-4.9	0.8	0.7	0.6	0.5		0.1	0.0	0.0	0.0	
contribution from real GDP growth	-24.4			-37.7	-28.7	-5.1	-5.0	-4.8	-4.3		-38.2	-7.7	0.4	0.0	
Contribution from real exchange rate depreciation	26.5			-22.3	-21.5	-1.0	-1.2	-0.5	-1.3		0.2	0.1	...	...	
Other identified debt-creating flows	-4.7			-4.3	-1.9	-1.6	-1.5	-1.4	-1.2		-0.3	-0.1	-0.2	-0.4	
Privatization receipts (negative)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Debt relief (HIPC and other)	-4.7			-4.3	-1.9	-1.6	-1.5	-1.4	-1.2		-0.3	-0.1	-0.2	-0.4	
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	0.0	
Residual, including asset changes	114.9			4.5	-307.1	-9.0	-3.6	-3.7	-2.4		-2.0	-2.0	-0.1	0.3	
NPV of public sector debt	168.1			25.5	26.4	27.3	27.7	27.7	26.2		7.1	2.9	2.9	5.7	
Of which: foreign-currency denominated	168.1			25.5	26.4	27.3	27.7	27.7	26.2		7.1	2.9	2.9	5.7	
external	168.1			25.5	26.4	27.3	27.7	27.7	26.2		7.1	2.9	2.9	5.7	
NPV of contingent liabilities (not included in public sector debt)	...		...		...	...	...	...	...		...	...	...	...	
Gross financing need 2/	-11.3			8.8	8.1	12.3	6.2	5.8	4.2		0.3	0.1	0.2	0.4	
NPV of public sector debt-to-revenue ratio (percent) 3/	181.2			34.8	45.5	52.5	49.8	52.0	47.5		46.6	39.4	18.0	13.1	
Of which: external	181.2			34.8	45.5	52.5	49.8	52.0	47.5		46.6	39.4	18.0	13.1	
Debt service-to-revenue ratio (percent) 3/ 4/	18.1			16.3	6.3	5.8	4.9	4.6	4.0		3.7	3.1	1.2	1.1	
Primary deficit that stabilizes the debt-to-GDP ratio	-105.1			67.5	364.1	16.0	10.7	9.8	8.8		38.5	7.9	-0.2	0.0	
Key macroeconomic and fiscal assumptions															
Nominal GDP (local currency)	765			983	1,261	1,496	1,750	2,016	2,292		8,968	22,732	26,758	18,505	
Real GDP growth (percent)	6.0	3.1	1.7	8.0	7.0	7.0	7.5	8.0	8.0	7.6	287.3	150.9	-9.9	-1.0	27.2
Average nominal interest rate on forex debt (percent)	1.2	1.5	1.1	1.2	0.8	3.2	3.1	3.0	2.9	2.4	2.9	3.0	3.3	1.6	2.8
Average real interest rate on domestic currency debt (percent)	...	5.7	14.1	...	...	...	...	...	...		...	...	...	...	
Real exchange rate depreciation (percent, + indicates depreciation)	6.6	1.6	6.6	-4.8	...	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, percent)	14.7	6.3	5.1	19.0	19.8	10.9	8.8	6.7	5.3	11.7	1.0	1.0	1.0	1.0	1.0
Growth of real primary spending (deflated by GDP deflator, percent)	-16.8	35.5	59.2	17.1	-4.4	4.7	3.6	3.7	8.9	5.6	1.7	23.3	5.4	4.7	13.6

Sources: São Tome and Principe authorities; and World Bank and IMF staff estimates and projections.

1/ Public sector debt refers to the gross central government debt.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues including grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations derived over the past 5 years.

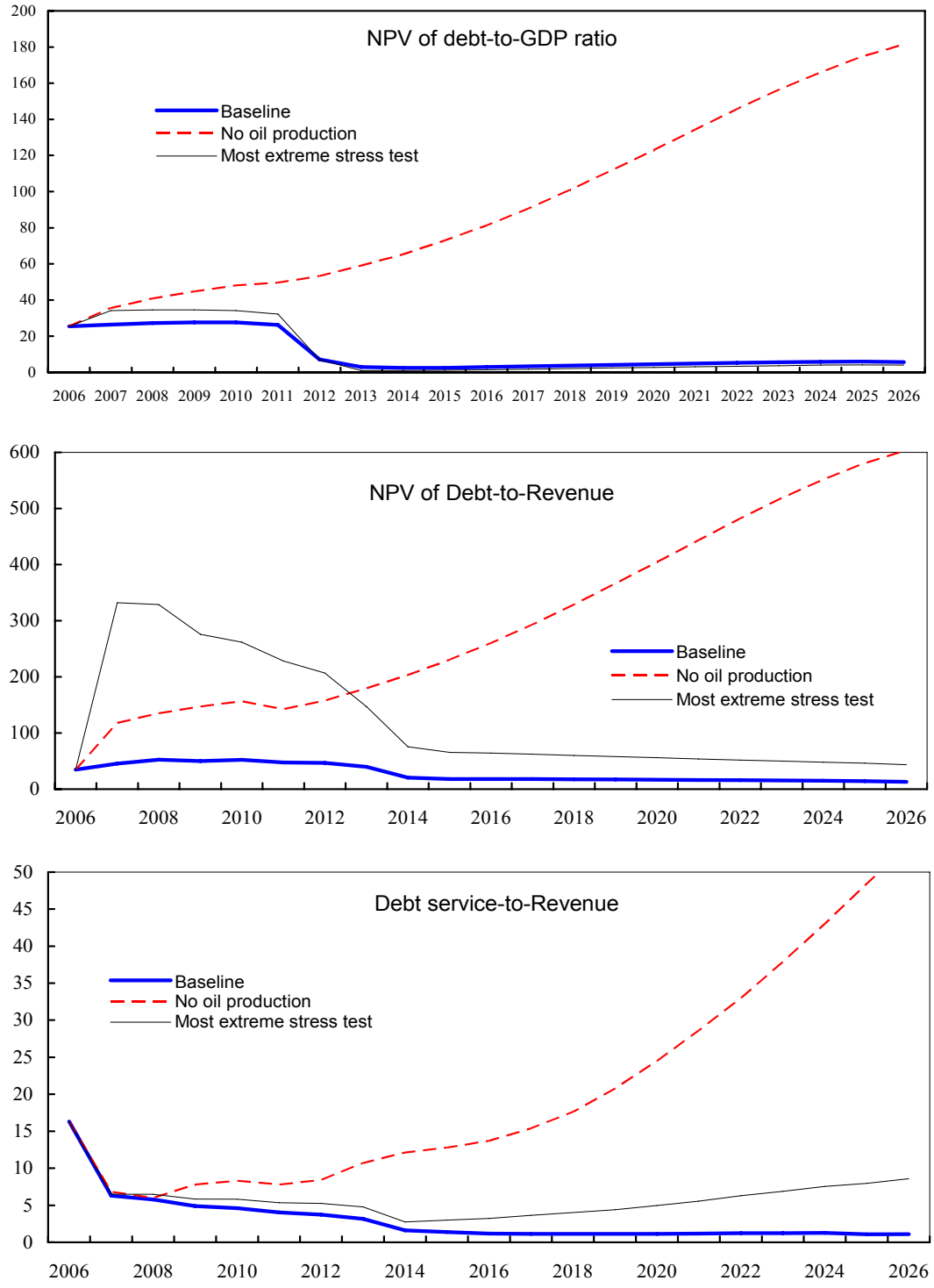
Table 2. São Tomé and Príncipe: Sensitivity Analysis for Key Indicators of Public Debt 2006–26
(Percent)

	Est.	Projections							
	2006	2007	2008	2009	2010	2012	2013	2020	2026
				NPV of Debt-to-GDP Ratio					
Baseline	25	26	27	28	28	7	3	5	6
A. Alternative scenarios									
A1. No oil production	25	36	41	45	48	53	59	123	182
A2. Primary balance is unchanged from 2007	25	26	23	25	26	9	6	44	89
A3. Permanently lower GDP growth 1/	25	27	28	29	29	6	1	8	24
A4. Topping up	25	21	22	24	25	7	3	4	6
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	25	30	35	38	41	10	3	21	54
B2. Primary balance is at historical average minus one standard deviations in 2007–08	25	42	53	52	51	11	3	5	7
B3. Combination of B1–B2 using one half standard deviation shocks	25	37	44	42	40	8	1	3	4
B4. One-time 30 percent real depreciation in 2007	25	193	171	153	139	32	11	15	19
B5. 10 percent of GDP increase in other debt-creating flows in 2007	25	34	35	35	34	7	1	3	4
				NPV of Debt-to-Revenue Ratio 2/					
Baseline	35	45	53	50	52	47	39	17	13
A. Alternative scenarios									
A1. No oil production	35	118	135	147	156	158	180	404	604
A2. Primary balance is unchanged from 2007	35	45	45	45	49	60	75	162	203
A3. Permanently lower GDP growth 1/	35	46	53	51	54	39	10	28	55
A4. Topping up	35	35	43	43	46	43	37	16	13
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	35	50	65	67	74	66	37	78	123
B2. Primary balance is at historical average minus one standard deviations in 2007–08	35	72	103	94	95	72	37	19	17
B3. Combination of B1–B2 using one half standard deviation shocks	35	63	82	73	72	48	13	10	9
B4. One-time 30 percent real depreciation in 2007	35	332	329	276	262	207	147	56	43
B5. 10 percent of GDP increase in other debt-creating flows in 2007	35	59	67	62	64	46	14	10	9
				Debt Service-to-Revenue Ratio 2/					
Baseline	16	6	6	5	5	4	3	1	1
A. Alternative scenarios									
A1. No oil production	16	7	6	8	8	8	11	24	54
A2. Primary balance is unchanged from 2007	16	6	6	4	4	4	5	10	15
A3. Permanently lower GDP growth 1/	16	6	6	5	5	4	3	2	4
A4. Topping up	16	5	4	4	4	3	3	1	1
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	16	7	6	6	6	5	5	5	9
B2. Primary balance is at historical average minus one standard deviations in 2007–08	16	6	9	8	6	5	5	2	2
B3. Combination of B1–B2 using one half standard deviation shocks	16	6	8	7	6	4	4	2	1
B4. One-time 30 percent real depreciation in 2007	16	8	8	7	6	5	4	2	2
B5. 10 percent of GDP increase in other debt-creating flows in 2007	16	6	7	5	5	4	4	1	1

Sources: São Tome and Principe authorities, and World Bank and IMF staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues including grants.

Fig. 1. São Tomé and Príncipe: Indicators of Public Debt, 2006–26 1/
(Percent)



Sources: World Bank and IMF staff projections.

1/ Baseline scenario includes HIPC and MDRI debt relief.

Table 3. São Tomé and Príncipe: External Debt Sustainability Framework, Baseline Scenario, 2006–26 1/
(Percent of GDP, unless otherwise indicated)

	Est. 2006	Historical Average 6/	Standard Deviation 6/	Projections 2007	2008	2009	2010	2011	2007–11 Average	2012	2013	2016	2026	2012–26 Average
External debt (nominal) 1/	**413.7**			**74.6**	**69.5**	**63.3**	**57.3**	**51.3**		**12.8**	**4.8**	**3.6**	**4.6**	
Of which : public and publicly guaranteed (PPG)	413.7			74.6	69.5	63.3	57.3	51.3		12.8	4.8	3.6	4.6	
Change in external debt	-38.5			-339.0	-5.1	-6.2	-6.0	-6.0		-38.5	-7.9	0.2	0.0	
Identified net debt-creating flows	-17.0			2.1	7.3	2.0	1.5	-1.1		-39.5	-32.3	-31.8	19.3	
Non-interest current account deficit	**56.9**	**23.9**	**16.3**	**61.0**	**62.7**	**62.5**	**62.8**	**230.2**		**49.3**	**-16.8**	**-19.5**	**25.8**	-4.2
Deficit in balance of goods and services	89.5			86.2	88.4	87.1	86.3	248.9		-3.6	-67.1	-55.5	-2.0	
Exports	28.5			26.4	25.9	25.1	24.1	23.3		82.0	94.4	91.3	54.3	
Imports	118.0			112.5	114.3	112.2	110.5	272.2		78.4	27.3	35.8	52.3	
Net current transfers (negative = inflow)	-34.4	-38.5	12.8	-26.7	-26.1	-25.1	-24.0	-19.2		-4.6	-1.7	-0.3	-0.4	-0.7
Other current account flows (negative = net inflow)	1.8			1.5	0.4	0.5	0.5	0.5		57.5	52.0	36.2	28.2	
Net FDI (negative = inflow)	**-34.6**	**-12.9**	**11.5**	**-34.2**	**-51.2**	**-56.5**	**-57.4**	**-227.8**		**-51.0**	**-7.9**	**-12.7**	**-6.6**	-9.2
Endogenous debt dynamics 2/	**-39.3**			**-24.7**	**-4.1**	**-4.0**	**-3.9**	**-3.5**		**-37.8**	**-7.6**	**0.4**	**0.1**	
Contribution from nominal interest rate	1.6			0.6	0.7	0.7	0.7	0.6		0.1	0.1	0.0	0.0	
Contribution from real GDP growth	-32.9			-25.3	-4.8	-4.7	-4.6	-4.1		-37.9	-7.6	0.4	0.0	
Contribution from price and exchange rate changes	-8.0			…	…	…	…	…		…	…	…	…	
Residual (3–4) 3/ 4/	**-21.5**			**-341.2**	**-12.4**	**-8.2**	**-7.5**	**-4.9**		**0.9**	**24.4**	**32.0**	**-19.3**	
NPV of external debt	18.4			21.0	23.1	23.9	24.5	24.5		6.7	2.8	2.8	6.0	
(percent of exports)	64.7			79.5	88.9	95.2	101.4	105.0		8.2	3.0	3.0	11.0	
NPV of PPG external debt	**18.4**			**21.0**	**23.1**	**23.9**	**24.5**	**24.5**		**6.7**	**2.8**	**2.8**	**6.0**	
In percent of exports	64.7			79.5	88.9	95.2	101.4	105.0		8.2	3.0	3.0	11.0	
Debt service-to-exports ratio (percent)	17.0			5.9	6.7	8.5	9.0	9.7		0.7	0.3	0.2	0.2	
PPG debt service-to-exports ratio (percent)	17.0			5.9	6.7	8.5	9.0	9.7		0.7	0.3	0.2	0.2	
Total gross financing need (billions of U.S. dollars)	0.0			0.0	0.0	0.0	0.0	0.0		0.0	-0.3	-0.5	0.2	
Non-interest current account deficit that stabilizes debt ratio	95.4			400.0	67.8	68.7	68.8	236.2		87.9	-8.9	-19.7	25.8	
Key macroeconomic assumptions														
Real GDP growth (percent)	8.0	4.7	1.7	7.0	7.0	7.5	8.0	8.0	7.5	287.3	150.9	-9.9	-1.0	27.2
GDP deflator in US dollar terms (percent change)	1.8	3.0	4.3	7.1	1.7	2.7	2.4	3.0	3.4	0.3	0.3	0.3	0.3	0.3
Effective interest rate (percent) 5/	0.4	0.1	0.2	0.2	1.0	1.1	1.2	1.2	0.9	1.1	1.0	0.8	0.4	0.6
Growth of exports of goods and services (U.S. dollar terms, percent)	7.0	4.8	6.9	6.0	7.0	6.8	6.4	7.4	6.7	1268.1	189.7	-12.0	-7.9	92.0
Growth of imports of goods and services (U.S. dollar terms, percent)	44.9	13.7	14.8	9.3	10.5	8.4	8.9	174.2	42.3	12.0	-12.4	16.1	4.8	4.0
Grant element of new public sector borrowing (including IMF; percent)	...	...	...	42.4	44.0	47.1	48.9	46.4	45.8	48.1	48.1	48.1	48.1	48.1
Grant element of new public sector borrowing (excluding IMF; percent)	...	...	...	46.9	47.6	49.3	48.9	46.4	47.8	48.1	48.1	48.1	48.1	48.1
Memorandum item:														
Nominal GDP (billions of U.S. dollars)	0.1			0.09	0.10	0.11	0.12	0.13		0.52	1.31	1.51	0.97	

Source: World Bank and IMF staff projections.

1/ External public debt only.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Includes deposits and withdrawals from the National Oil Account.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations for the 1999–05 period.

Table 4. São Tomé and Príncipe: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006–26
(Percent)

	Est.	Projections						
	2006	2007	2008	2009	2010	2013	2020	2026
				NPV of debt-to-GDP ratio 1/				
Baseline	18	21	23	24	24	3	4	6
A. Alternative Scenarios								
A1. No oil exports 2/	18	46	49	51	52	37	101	166
A2. New public sector loans on less favorable terms in 2007–26 3/	18	21	25	29	30	4	7	10
A3. Topping up	18	15	18	20	21	3	4	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	18	30	34	34	34	4	5	7
B2. Export value growth at historical average minus one standard deviation in 2007–08 4/	18	22	24	25	25	3	4	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007–08	18	36	40	40	40	4	6	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007–08 5/	18	35	36	36	36	4	4	3
B5. Combination of B1–B4 using one-half standard deviation shocks	18	36	40	41	40	5	6	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	18	29	32	79	77	8	10	12
				NPV of debt-to-exports ratio				
Baseline	65	79	89	95	101	3	6	11
A. Alternative Scenarios								
A1. No oil exports 2/	65	173	187	193	198	119	297	452
A2. New public sector loans on less favorable terms in 2007–26 3/	65	79	95	114	125	4	8	18
A3. Topping up	65	57	69	78	87	3	5	11
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	65	114	116	122	127	4	6	12
B2. Export value growth at historical average minus one standard deviation in 2007–08 4/	65	91	111	118	125	4	6	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007–08	65	138	134	139	144	4	7	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007–08 5/	65	132	140	143	148	4	5	5
B5. Combination of B1–B4 using one-half standard deviation shocks	65	128	145	150	156	4	7	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	65	110	89	227	230	6	10	15
				Debt service-to-exports ratio				
Baseline	17.0	5.9	6.7	8.5	9.0	0.3	0.1	0.2
A. Alternative Scenarios								
A1. No oil exports 2/	17.0	12.6	13.6	17.2	18.2	12.6	22.7	44.6
A2. New public sector loans on less favorable terms in 2007–26 3/	17.0	5.7	6.7	8.9	10.1	0.3	0.2	0.5
A3. Topping up	12.2	3.8	4.1	6.2	6.8	0.3	0.1	0.1
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007–08	17.0	6.2	7.0	8.8	9.3	0.3	0.2	0.3
B2. Export value growth at historical average minus one standard deviation in 2007–08 4/	17.0	6.4	8.0	10.3	11.0	0.4	0.2	0.3
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2007–08	17.0	6.5	7.2	9.0	9.5	0.3	0.3	0.4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007–08 5/	17.0	5.9	7.3	10.1	10.5	0.3	0.5	0.9
B5. Combination of B1-B4 using one-half standard deviation shocks	17.0	6.2	8.0	10.0	10.6	0.3	0.3	0.5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 6/	17.0	8.2	8.4	10.2	10.6	0.3	0.5	0.9
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 7/	53	51	49	47	45	38	33	32

Source: World Bank and IMF staff projections.

1/ By the year 2023 real GDP growth slows down significantly due to the projected decline in petroleum production.

2/ Variables include real GDP growth, growth of GDP deflator (U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

3/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

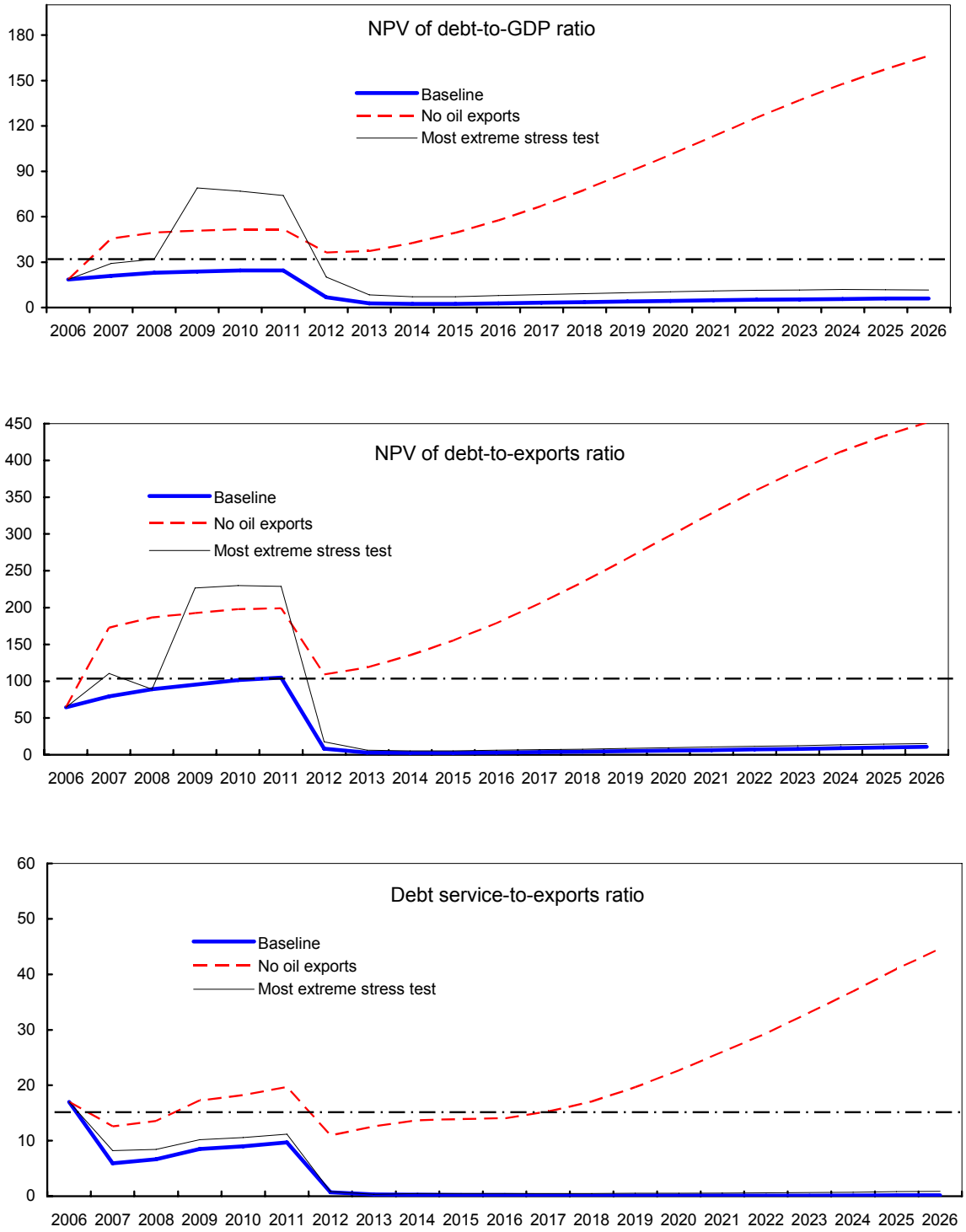
4/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

5/ Includes official and private transfers and FDI.

6/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

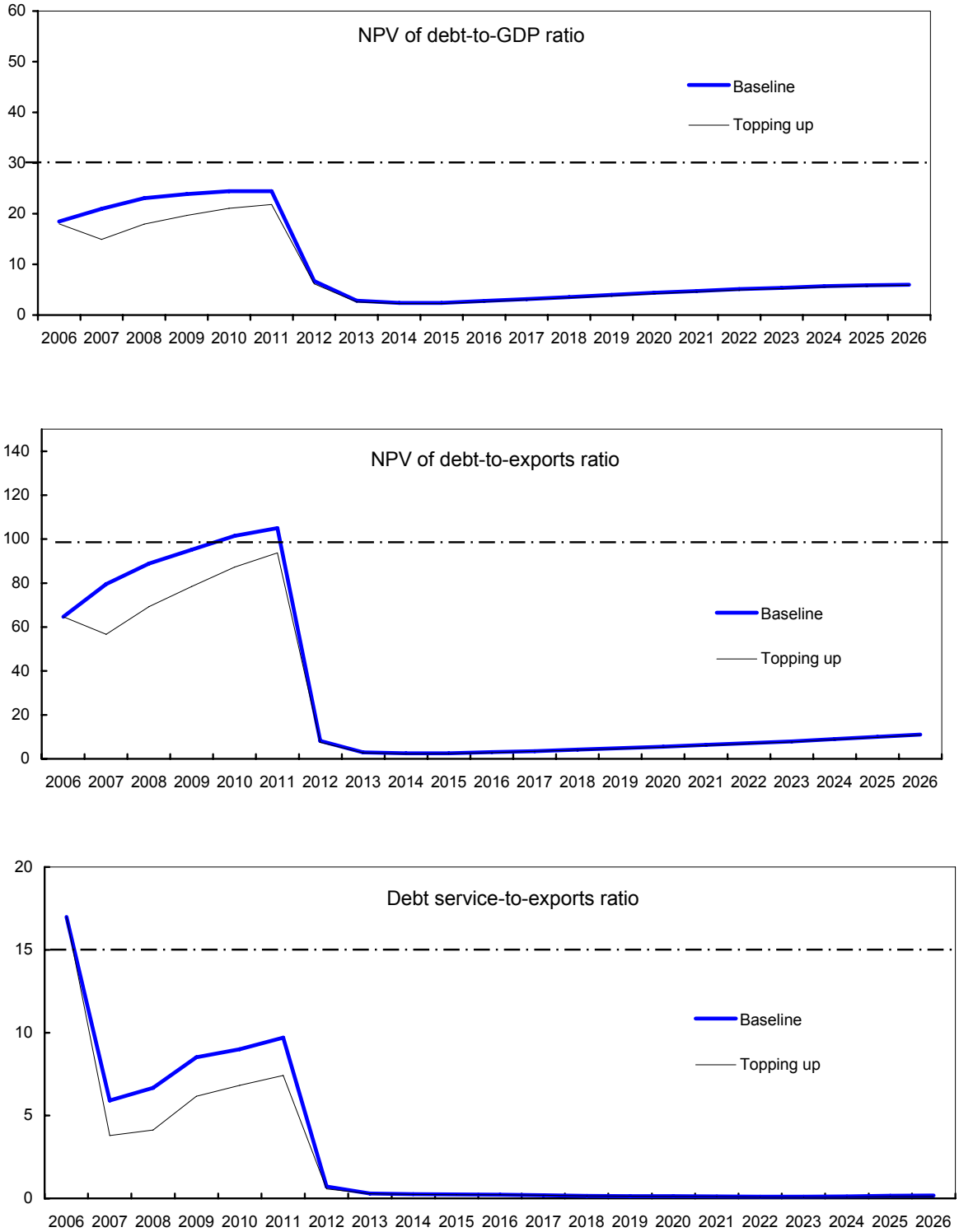
7/ Based on historical level of concessionality. Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Fig. 2. São Tomé and Príncipe: Indicators of External Debt, 2006–26 1/
(Percent)



Source: World bank and IMF staff projections.
1/ Baseline scenario includes HIPC and MDRI debt relief.

Fig. 3. São Tomé and Príncipe: Indicators of External Debt, 2006–26 1/
(Percent)



Source: World Bank and IMF staff projections.
1/ Baseline scenario includes HIPC and MDRI debt relief.